Topic 1: Basic Concepts

CSE 4405: Data and Telecommunications

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Part I: Introduction

Learning objectives

- Define data communication and its effectiveness criteria
- Identify system components and data-flow modes
- Compare network types, structures, and topologies
- Explain the Internet, protocols, and standards

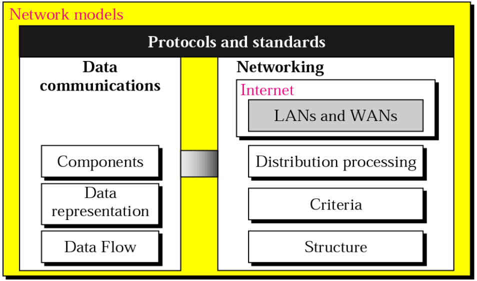
Roadmap

- Data communications
- Networks and internetworks
- The Internet
- Protocols and standards

Data Communications and Networking

Focus of this chapter

We begin with the language of networking: data representation, communication between devices, network organization, and the role of protocols and standards.



- Start from communication between two devices
- Expand to network structures and the Internet
- End with the protocols and standards that enable interoperability

1.1 Data Communications

Definition

Telecommunication means communication at a distance. **Data communication** is the exchange of data between two devices through a transmission medium such as copper wire, fiber, or radio.

In this section

- Characteristics of effective communication
- Components of a communication system
- Data representation
- Direction of data flow

Fundamental Characteristics of Data Communication

- **Delivery**: data must reach the correct destination.
- **Accuracy**: data must arrive without uncorrected errors.
- **Timeliness**: data must arrive when needed; delayed delivery may be useless.
- **Jitter**: variation in arrival time should be low, especially for audio and video.

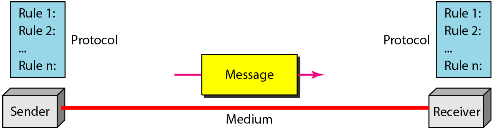
Lecture view

A system is not "good" merely because it connects two devices. It must deliver the right data, correctly, and at the right time.

Five Components of a Data Communications System

- **Message**: the information being communicated
- **Sender**: the device that originates the message
- **Receiver**: the device that accepts the message
- **Medium**: the path used to carry the signal
- **Protocol**: the rules governing communication

Communication needs both a transmission path and agreed rules.



Components in Plain Language

Component	Meaning	Typical examples
Message	The data to be sent	Text, numbers, images, audio, video
Sender	The source device	Computer, phone, camera, workstation
Receiver	The destination device	Computer, display, phone, storage server
Medium	The transmission path	Twisted pair, coaxial, fiber, radio
Protocol	Shared communication rules	Format, timing, addressing, acknowledgments

Why this matters

A cable alone is not enough. Two connected devices still need a common protocol to interpret the exchanged signals correctly.

Data Representation: Text and Numbers

- **Text** is represented as bit patterns using character codes.
- Common coding ideas include **ASCII**, **extended ASCII**, and **Unicode**.
- **Numbers** are also represented as bit patterns, but the encoding rules differ from those for text.
- The receiver must know the same representation scheme used by the sender.

Example

The letter `A` and the number 65 may be stored as bits, but they are interpreted differently because their encoding rules are different.

Data Representation: Images, Audio, and Video

- **Images**: represented as picture elements with encoded values.
- **Audio**: naturally continuous, but digitized for storage and transmission.
- **Video**: a sequence of images, often combined with audio.
- Different data forms need different encoding and compression methods.

Teaching point

The same network can carry text, sound, and video because all of them can ultimately be represented in bits.

Direction of Data Flow

- **Simplex**: communication is one-way only.
- **Half-duplex**: both ends can send and receive, but not simultaneously.
- **Full-duplex**: both ends can send and receive at the same time.

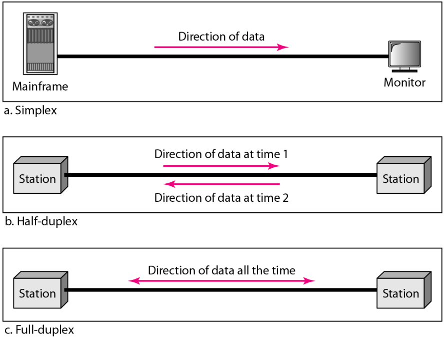
Simple examples

Keyboard to monitor is a simplex-style example. Walkie-talkies are half-duplex. A phone call is full-duplex.

Data Flow Modes in a Figure

Figure: Simplex, half-duplex, and full-duplex data flow

- The figure compares how information moves between two endpoints.
- The key difference is whether both sides may transmit simultaneously.
- This choice affects efficiency, design complexity, and suitable applications.



1.2 Networks

Network definition

A network is a set of interconnected devices, called **nodes**, linked by communication channels. A node may be a computer, printer, router, or any device that can send or receive data.

In this section

- Distributed processing
- Network criteria
- Physical structures and topologies
- Categories of networks and internetworks

Distributed Processing

- A task is divided among multiple computers instead of being handled by one large machine.
- Each device performs a subset of the overall work.
- This improves scalability, specialization, and resource sharing.

Real-world example

In an online shopping system, one server may handle the web interface, another may store product data, and another may process payments.

Network Criteria

Criterion	What it asks
Performance	How fast and efficiently does the network deliver data?
Reliability	How often does it fail, and how quickly does it recover?
Security	How well does it protect data and services from misuse or attack?

These three criteria provide a practical way to evaluate any network design.

Performance

- Performance depends on the number of users and the load they generate.
- The transmission medium and its data rate strongly affect throughput.
- Hardware and software quality influence both speed and error-free delivery.
- Two common metrics are **throughput** and **delay**.

Interpretation

A high-throughput network is not always satisfactory if the delay is too large for interactive or real-time traffic.

Reliability and Security

Reliability

- Frequency of failure
- Time needed to recover
- Protection against catastrophic events

Security

- Protection from unauthorized access
- Protection from malware such as viruses and worms
- Protection of confidentiality, integrity, and availability

Concrete view

A fast network is still poor if it fails often or exposes sensitive data.

Physical Structures

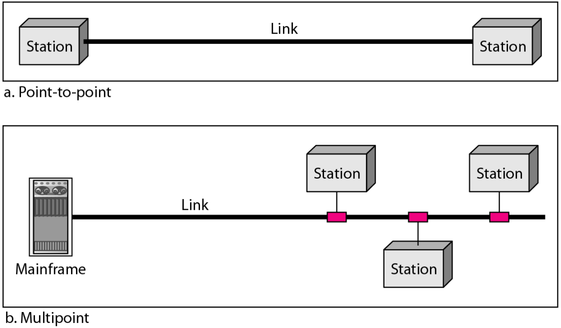
Two basic connection types

A physical structure describes how devices share links. The first distinction is between **point-to-point** and **multipoint** connections.

- **Point-to-point**: one dedicated link connects exactly two devices.
- **Multipoint** or **multidrop**: multiple devices share the same link.

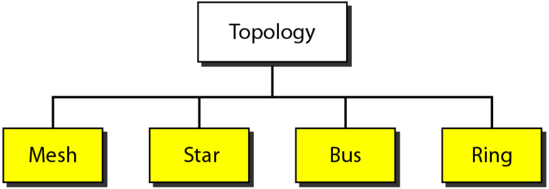
Point-to-Point and Multipoint Connections

Figure: Types of physical connections



- Point-to-point links are dedicated and simple to reason about.
- Multipoint links are shared and therefore need coordination among users.
- The choice affects efficiency, cost, and control.

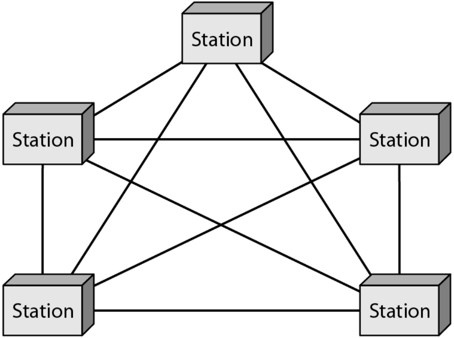
Figure: Common network topologies



- Topology describes the arrangement of nodes and links.
- The most common forms are mesh, star, bus, ring, tree, and hybrid.
- Each topology makes a different tradeoff among cost, robustness, and manageability.

Mesh Topology

- Every device has a dedicated point-to-point link to every other device.
- A fully connected mesh with n devices needs
$$\frac{n(n-1)}{2}$$
links.
- Mesh provides strong isolation and redundancy.



Mesh Topology: Advantages and Limitations

Advantages

- Dedicated links carry their own traffic
- Easy fault isolation
- Good privacy and robustness

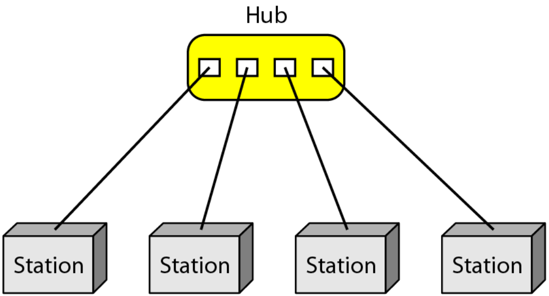
Limitations

- Large amount of cabling
- Many I/O ports per device
- High cost and difficult expansion

Star Topology

- Each device connects only to a central hub or switch.
- All traffic passes through that central controller.
- This makes installation and management straightforward.

The failure of one outer link affects one device only, but failure of the central device affects the whole star.



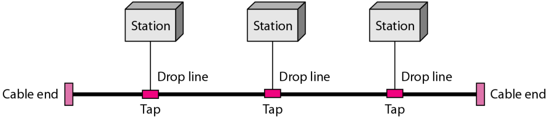
Tree Topology

- A tree topology is a hierarchical variation of the star topology.
- An **active hub** contains a repeater and regenerates signals before forwarding them.
- A **passive hub** provides only a physical connection among attached devices.
- Tree structures are useful when a network grows in layers or branches.

Bus Topology

- A bus is a multipoint topology built around one shared backbone cable.
- Devices connect through **drop lines** and **taps**.
- All stations share the same main transmission path.

A bus is simple and economical, but the shared backbone becomes a critical point of failure.



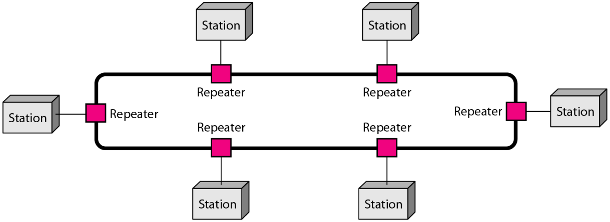
Bus and Ring Topology: Short Comparison

Bus

- Easy to install
- Shared backbone lowers cost
- Reconfiguration and fault isolation are harder

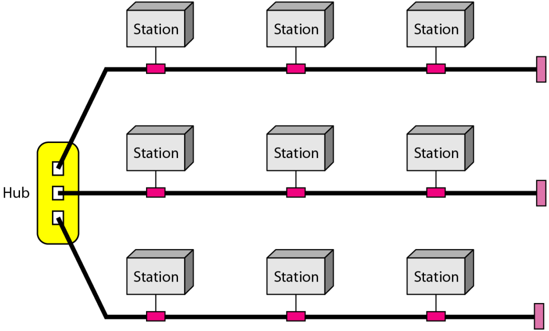
Ring

- Each device connects to two neighbors
- Simple fault localization
- A single break can disable the ring; dual rings improve resilience



Hybrid Topology

- Real networks often combine topologies rather than using only one.
- A common example is a star backbone with smaller bus-style or star-style branches.
- Hybrid designs are chosen to match scale, cost, and local constraints.



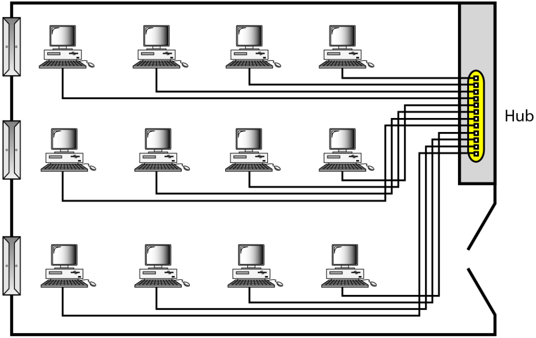
Categories of Networks

Type	Typical scope	Ownership	Typical use
LAN	Room to campus	Usually private	Local resource sharing
MAN	City or metropolitan area	Often provider-managed	Urban interconnection
WAN	Country to world	Carrier or provider infrastructure	Long-distance communication

The three categories differ mainly in scale, management, and the distance over which they operate.

LAN: Local Area Network

- A LAN is usually privately owned.
- It connects devices in a room, building, or campus.
- LANs support fast local sharing of files, printers, and applications.

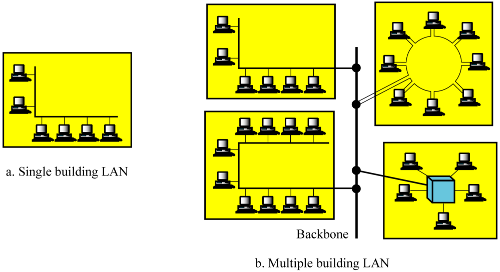


Use case

A university lab or office floor typically uses a LAN to connect PCs, printers, and local servers.

LAN and MAN in Figures

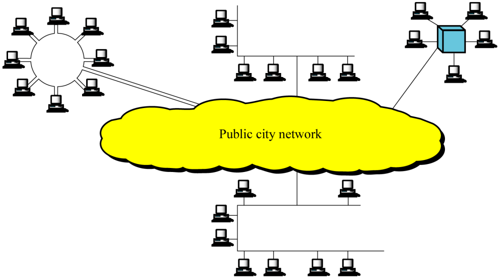
Figure: Extended LAN



A

LAN can extend beyond one room or one floor while still remaining local in scope.

Figure: Metropolitan Area Network



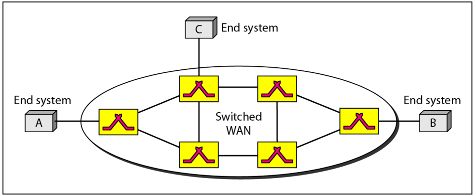
A

MAN interconnects local networks across a city or metropolitan region.

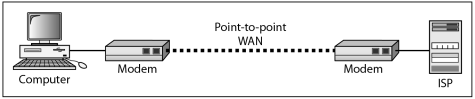
WAN: Wide Area Network

- A WAN provides long-distance transmission of data, voice, image, and video.
- It may span a country, a continent, or the whole world.
- WANs rely on intermediate provider infrastructure and switching equipment.

The figure illustrates both switched WANs and point-to-point WAN links.



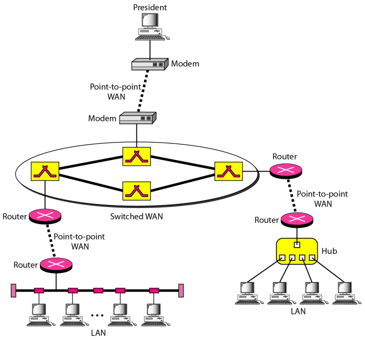
a. Switched WAN



b. Point-to-point WAN

Interconnection of Networks: Internetwork

- When two or more networks are connected, they form an **internetwork**.
- The **Internet** is the best-known example of a global internetwork.
- Routers make communication possible across otherwise separate networks.



Example

A university LAN becomes part of an internetwork when it connects through an ISP to other networks.

1.3 The Internet

Why this topic matters

The Internet changed how we learn, communicate, work, and do business by connecting independent networks into one global communication system.

- It is not one single network; it is a network of networks.
- It grew from research networking into a global public infrastructure.
- Understanding its history explains why protocols and ISPs matter.

Internet History: From ARPANET to TCP/IP

- **1967**: ARPA presented the idea of ARPANET.
- **1969**: ARPANET became operational with four nodes: UCLA, UCSB, SRI, and the University of Utah.
- **1972**: Vint Cerf and Bob Kahn launched the Internetting Project.
- **1973**: Cerf and Kahn described the ideas that led to TCP/IP.
- **1983**: TCP/IP became the official protocol suite for ARPANET.
- **1983**: MILNET separated military traffic from the research network.

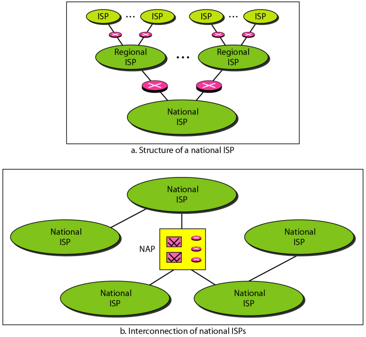
Internet History: Expansion and Commercialization

Year	Milestone
1977	An internet was tested using TCP/IP
1978	UNIX spread to academic sites
1986	NSFNET was established; the first IETF meeting was held
1990	ARPANET was replaced by NSFNET
1991	The World Wide Web emerged from CERN
1995	NSFNET became a research network and commercial ISPs expanded public access
2006	GENI proposed a platform for future Internet experimentation

The history is easier to read as stages: research origin, protocol standardization, academic expansion, then public and commercial growth.

Internet Hierarchy Today

Figure: Hierarchical organization of the Internet



- Users connect through local or access networks.
- Regional, national, and international providers interconnect those networks.
- This hierarchy makes the Internet scalable and economically manageable.

1.4 Protocols and Standards

Main question

How can devices from different vendors and networks communicate correctly? The answer is: they must follow common protocols and standards.

- Protocols define communication rules.
- Standards make those rules widely accepted and interoperable.
- Standards organizations coordinate the process.

What Is a Protocol?

- A **protocol** is a set of rules governing data communication.
- Communication occurs between **entities** in different systems.
- An entity may be any component capable of sending or receiving information.
- Shared rules are necessary if different devices are to interpret the same signals in the same way.

Simple example

If one device sends an acknowledgment after every frame and the other does not expect it, communication breaks down.

The Three Key Elements of a Protocol

Element	Meaning	Simple example
Syntax	Structure or format of the data	Field order in a frame header
Semantics	Meaning of each field and required action	Meaning of an ACK or error flag
Timing	When data are sent and at what rate	Send every 20 ms or retransmit after timeout

Why Standards Matter

- Standards make products from different manufacturers work together.
- They support open markets and large-scale interoperability.
- **De facto** standards become common through widespread use.
- **De jure** standards are formally approved by recognized bodies.

Related terms

By fact is another way to describe de facto adoption; **by law** refers to officially approved standards.

Open, Proprietary, and Formal Standards

- **Proprietary** standards are created by a company for its own products.
- **Nonproprietary** or **open** standards are publicly available.
- Open standards are especially important when many vendors must interoperate.
- Formal approval usually improves trust, adoption, and compatibility.

Standards Organizations: ISO and ITU-T

- **ISO**: the **International Organization for Standardization**, established in 1947.
- ISO is a multinational body devoted to international agreement on standards.
- **ITU-T**: the **International Telecommunication Union Telecommunication Standardization Sector**.
- ITU-T develops standards for global telecommunications.
- Well-known ITU-T families include the **V-series** and **X-series**.

Standards Organizations: ANSI, IEEE, and EIA

- **ANSI**: the **American National Standards Institute**; it coordinates standards activity in the United States.
- **IEEE**: the **Institute of Electrical and Electronics Engineers**; a major standards source for computing and communications.
- **EIA**: historically associated with interface standards such as EIA-232 and EIA-530.
- Together, these bodies help make heterogeneous systems interoperable.

Forums, Regulators, and National Examples

Forums and regulators

- Forums evaluate and promote emerging technologies quickly.
- Examples include the **Frame Relay Forum** and the **ATM Forum**.
- The **FCC** regulates communications in the United States.

Selected national examples

- **TTA**: Telecommunications Technology Association
- **KATS**: Korean Agency for Technology and Standards

http://www.tta.or.kr
http://www.kats.go.kr

Internet Standards and RFCs

- The **IETF** (Internet Engineering Task Force) is central to Internet standardization.
- A specification typically begins as an **Internet-Draft**.
- An Internet-Draft is a working document with no official status.
- **RFCs** (**Requests for Comments**) publish technical proposals, specifications, and standards-related material.

Clean interpretation

Not every RFC is a standard, but the RFC series is the main publication channel for Internet protocol work.

Part 1 Summary

- Effective data communication depends on delivery, accuracy, timeliness, and low jitter.
- A communication system needs a message, sender, receiver, medium, and protocol.
- Data may flow in simplex, half-duplex, or full-duplex mode.
- Networks are evaluated by performance, reliability, and security.
- Common topologies include mesh, star, bus, ring, tree, and hybrid.

Part 1 Summary (cont'd)

- LANs, MANs, and WANs differ mainly in scale and management.
- An internetwork connects multiple networks; the Internet is the best-known example.
- Protocols define syntax, semantics, and timing.
- Standards support interoperability and are developed by organizations such as ISO, ITU-T, ANSI, IEEE, EIA, and the IETF.
- RFCs and Internet-Drafts are central to Internet standards work.

Part II: The OSI Model

Learning objectives

- Explain why layered models are useful
- Describe the OSI model and peer processes
- Compare OSI and TCP/IP
- Distinguish physical, logical, port, and specific addresses

2.1 Layered Tasks

Why layering?

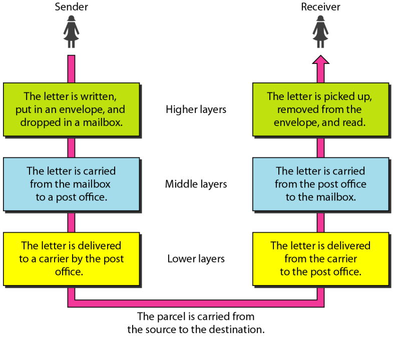
Complex communication becomes manageable when responsibilities are divided into layers. Each layer solves a limited part of the overall problem.

- Everyday communication systems also use layers
- Layers provide services to the layer above
- Layers rely on services from the layer below

Layered Tasks: Sender, Receiver, and Carrier

Figure: Layered roles in a communication task

- The figure uses a postal-mail analogy.
- Different parties handle message creation, transport, and delivery.
- Networking layers work the same way: not every component does every job.



Hierarchy and Services

- A layered system has **higher**, **middle**, and **lower** responsibilities.
- Each layer uses the service of the layer immediately below it.
- Each layer hides unnecessary implementation details from the layers above it.
- This structure improves modularity and makes design changes easier.

Concrete view

An application should not need to know how individual electrical or radio signals are produced on the medium.

2.2 The OSI Model

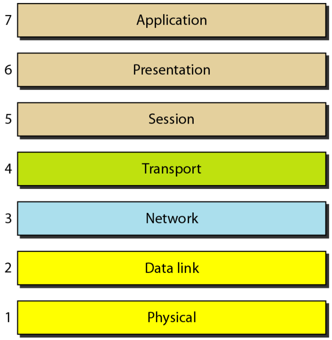
ISO and OSI

The **International Organization for Standardization (ISO)** developed the **Open Systems Interconnection (OSI)** reference model to organize communication functions into layers. ISO is the organization; OSI is the model.

- The OSI model is a reference framework, not one single protocol
- It groups related networking tasks into seven layers
- It helps explain interoperability among different systems

Layered Architecture of the OSI Model

- The seven layers are Physical, Data Link, Network, Transport, Session, Presentation, and Application.
- Layers 1–3 provide **network support**.
- Layers 5–7 provide **user support**.
- Layer 4, Transport, connects the two groups.

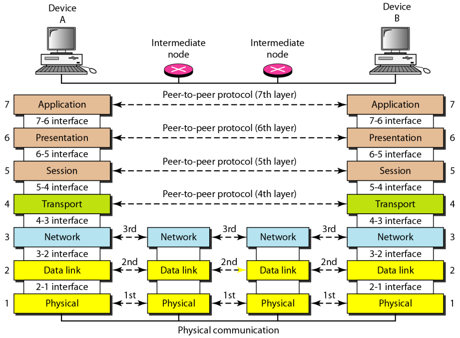


Peer-to-Peer Processes

- Layer *n* on one machine communicates logically with layer *n* on another machine.
- These logical relationships are called **peer-to-peer processes**.
- Each interface defines the services that a layer offers to the layer above it.
- Clear interfaces make the network architecture modular and replaceable.

Interaction Between Layers

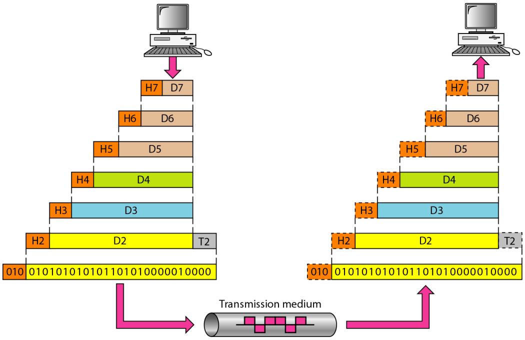
Figure: Interaction between layers in the OSI model



- A layer does not send data directly downward across the network.
- Instead, it hands data to the next lower local layer.
- Peer communication is logical; actual transmission happens through lower layers.

Encapsulation in Steps

Figure: An exchange using the OSI model



1. An upper layer creates data.
2. Each lower layer adds its own control information.
3. The receiver removes these headers in reverse order.

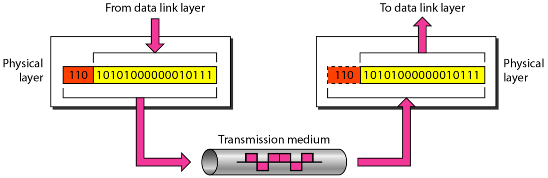
This wrapping process is called **encapsulation**.

2.3 Layers in the OSI Model

L	Layer	Main role
7	Application	Services for user applications
6	Presentation	Translation, compression, encryption
5	Session	Dialog control and synchronization
4	Transport	Process-to-process delivery
3	Network	Source-to-destination packet delivery
2	Data Link	Hop-to-hop frame delivery
1	Physical	Transmission of bits over the medium

Physical Layer

- The physical layer transmits a bit stream over the medium.
- It handles movement of individual bits from one node to the next.
- It deals with interfaces, signaling, and transmission characteristics.



Physical Layer: Main Concerns

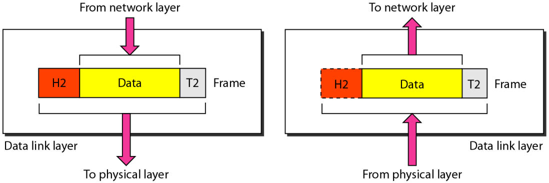
- Physical characteristics of interfaces and media
- Representation and synchronization of bits
- Data rate and transmission mode
- Line configuration and physical topology

Takeaway

The physical layer cares about how bits are carried, not about the meaning of those bits.

Data Link Layer

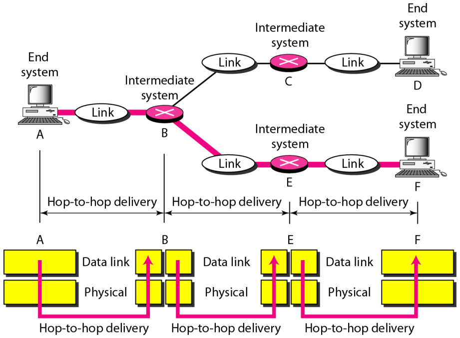
- The data link layer moves **frames** from one hop to the next.
- It provides node-to-node delivery over a single link.
- It sits above the physical layer and uses the medium prepared below it.



Data Link Layer: Duties and Hop-to-Hop Delivery

Major duties

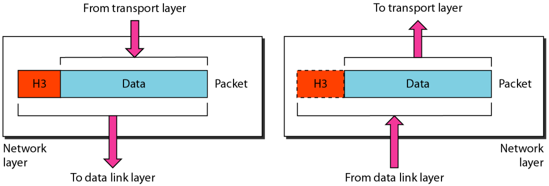
- Framing
- Physical addressing
- Flow control
- Error control
- Access control



The data link layer solves delivery on one link at a time, not across the entire internetwork.

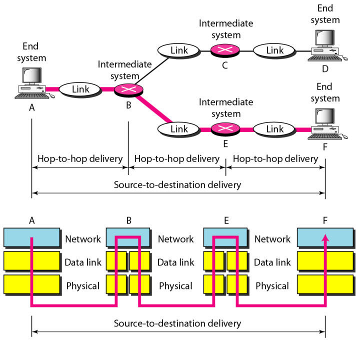
Network Layer

- The network layer delivers packets from the source host to the destination host.
- It is responsible for communication across multiple interconnected networks.
- Its main concerns are logical addressing and routing.



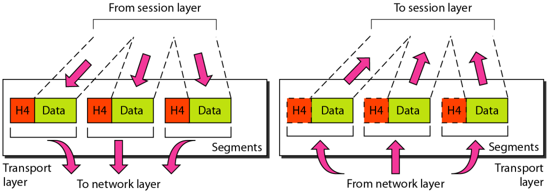
Network Layer: Addressing and Routing

- **Logical addressing** identifies hosts in an internetwork.
- **Routing** selects a path across multiple networks.
- The network layer solves source-to-destination delivery.



Transport Layer

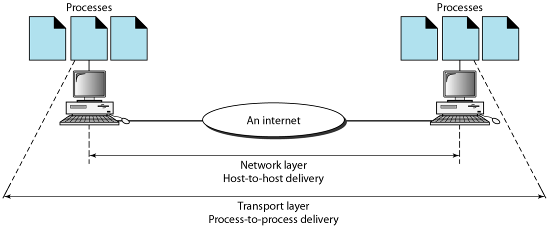
- The transport layer delivers a complete message from one process to another.
- It provides end-to-end process-to-process delivery.
- It sits above the network layer and below the session layer.



Transport Layer: Reliability and Control

Main responsibilities

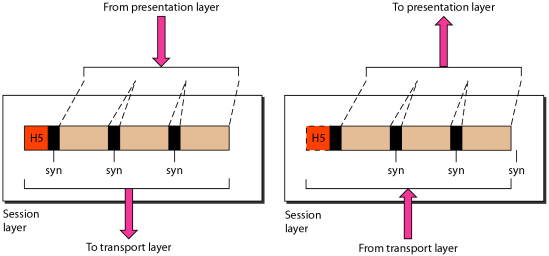
- Port addressing
- Segmentation and reassembly
- Connection control
- Flow control and error control



The transport layer is the first layer whose job is defined end to end for applications, not hop by hop for links.

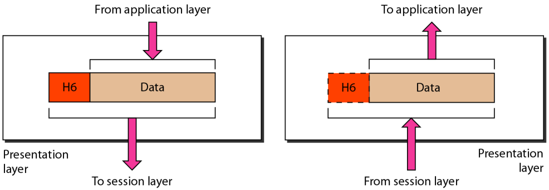
Session Layer

- The session layer manages dialog between communicating applications.
- It also provides synchronization points for long exchanges.
- It helps structure who speaks when and how a session resumes after interruption.



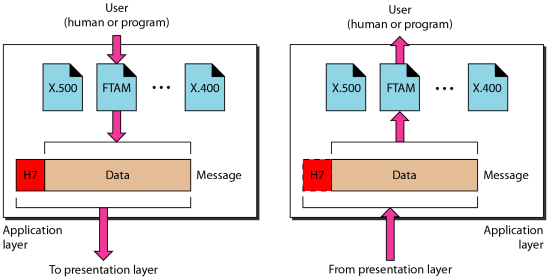
Presentation Layer

- The presentation layer handles translation, compression, and encryption.
- It ensures that data have a representation the receiver can interpret.
- It is concerned with meaning-preserving transformation of data.



Application Layer

- The application layer provides services directly to user applications.
- It is the closest OSI layer to the end user.
- Typical services include file transfer, mail, and directory access.



Application Layer: Typical Services

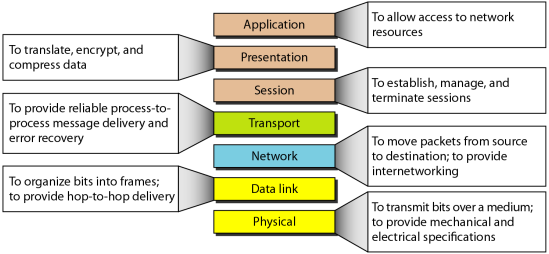
- Network virtual terminal
- File transfer, access, and management
- Mail services
- Directory services

Lecture view

When users talk about "using the network," they usually mean using application-layer services built on all lower layers.

Summary of the OSI Layers

Figure: Responsibilities across the OSI stack



- Lower layers are closer to the medium.
- Upper layers are closer to the user and the application.
- Each layer solves a different communication problem.

2.4 TCP/IP Protocol Suite

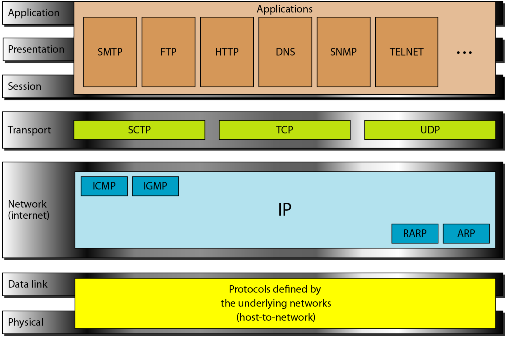
Practical counterpart

OSI is the main conceptual model. TCP/IP is the practical protocol suite used on the Internet.

- TCP/IP is often shown as a four-layer model historically
- In comparison with OSI, it is commonly discussed as a five-layer model
- It includes physical, data link, network, transport, and application layers

TCP/IP Compared with OSI

Figure: TCP/IP and OSI models



- TCP/IP combines the OSI session, presentation, and application layers into one application layer.
- OSI is more detailed as a reference model.
- TCP/IP is more directly tied to deployed Internet protocols.

TCP/IP Physical and Data Link Layers

- TCP/IP does not define one single protocol for the physical and data link layers.
- Instead, it works with many standard and proprietary lower-layer technologies.
- This is why a TCP/IP internetwork can include both LAN and WAN technologies.

Interpretation

TCP/IP is flexible about the local network technology as long as IP can operate above it.

TCP/IP Network Layer Protocols

- **IP** (**Internet Protocol**) provides logical addressing and packet delivery across networks.
- **ARP** resolves an IP address to a physical address on a local network.
- **RARP** performs the reverse lookup in legacy settings.
- **ICMP** carries control and error-reporting messages.
- **IGMP** (**Internet Group Management Protocol**) manages multicast group membership.

TCP/IP Transport Layer

- IP provides host-to-host delivery.
- **TCP**, **UDP**, and **SCTP** provide process-to-process delivery.
- **TCP** emphasizes reliability and ordered delivery.
- **UDP** emphasizes simplicity and low overhead.
- **SCTP** supports message-oriented transport with additional control features.

TCP/IP Application Layer

- The TCP/IP application layer combines the OSI session, presentation, and application layers.
- It includes many user-facing Internet protocols.
- Typical examples include web access, e-mail, file transfer, and name resolution.

Examples

HTTP, SMTP, FTP, and DNS are all application-layer protocols in the TCP/IP model.

2.5 Addressing

Why multiple address types?

A message must reach the correct network, the correct host, the correct interface on each hop, and finally the correct application process. One address type is not enough for all of these tasks.

- Physical addresses
- Logical addresses
- Port addresses
- Specific addresses

Address Types in TCP/IP

Figure: Addresses in TCP/IP

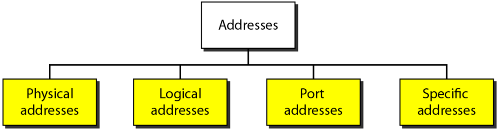
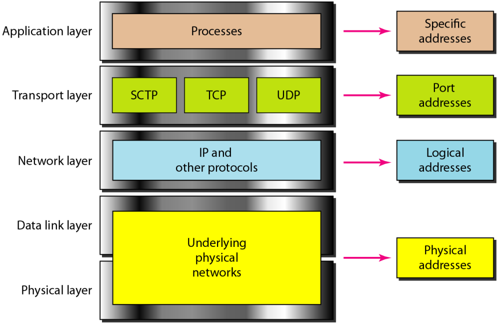


Figure: Relationship of layers and addresses



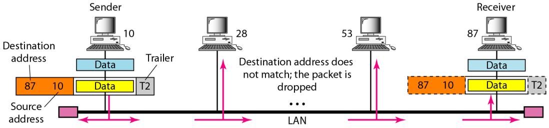
Different layers use different addresses because they solve different delivery problems.

Physical Addresses

- A **physical address**, also called a **link address**, is defined by the LAN or WAN technology in use.
- It appears in the frame created by the data link layer.
- It has authority only on the current local network or hop.
- Its format depends on the network technology.

Physical Addresses: Example 2.1

- A node with physical address 10 sends a frame to a node with physical address 87.
- Both nodes are on the same local link.
- The address is meaningful only within that local network.



Physical Addresses: Example 2.2

- Many LANs use a 48-bit, or 6-byte, physical address.
- It is commonly written as 12 hexadecimal digits separated by colons.
- This format is familiar from Ethernet MAC addresses.

Example format

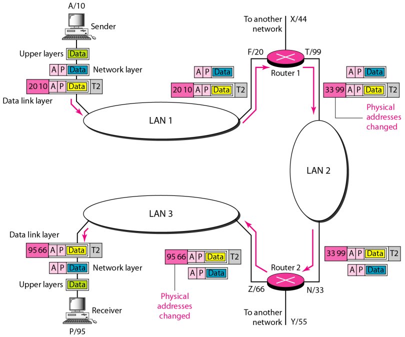
`07:01:02:01:2C:4B`

Logical Addresses

- Logical addresses are necessary for communication across an internetwork.
- Different physical networks use different local address formats.
- A universal logical address lets hosts be identified consistently across heterogeneous networks.

Logical Addresses: Example 2.3

- Hosts usually have one logical/physical address pair per interface.
- Routers have multiple address pairs because they connect multiple networks.
- Physical addresses change hop by hop, but logical addresses usually remain the same end to end.

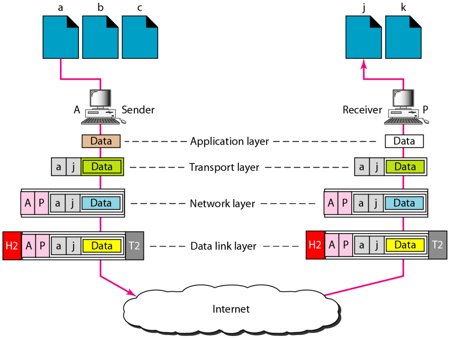


Port Addresses

- Physical and logical addresses deliver data to the correct host.
- Communication on the Internet ultimately occurs between **processes**.
- A **port address** identifies the specific process that should receive the data.
- In TCP/IP, a port address is 16 bits long.

Port Addresses: Example 2.4

- One host may run many processes at the same time.
- Port addresses distinguish among those processes.
- Physical addresses may change hop by hop, but logical and port addresses remain stable end to end.



Port Addresses and Specific Addresses

Example 2.5

- Port addresses are usually written as decimal numbers.
- They identify service endpoints or application processes.

Port format

753

Specific addresses

- User-friendly application-level identifiers
- Examples: e-mail addresses and **URLs** (**Uniform Resource Locators**)

Part 2 Summary

- Layered models divide complex communication into manageable responsibilities.
- The OSI model has seven layers, from the physical medium up to user applications.
- Peer-to-peer communication is logical; actual transmission depends on lower local layers.
- Encapsulation adds control information as data move downward through the stack.
- Physical, data link, and network layers mainly support the network; upper layers support users and applications.

Part 2 Summary (cont'd)

- TCP/IP is the practical protocol suite used on the Internet.
- Its lower layers are flexible about local network technology.
- IP, ARP, RARP, ICMP, and IGMP support network-layer operation in TCP/IP.
- TCP, UDP, and SCTP provide transport services to application processes.
- Communication in TCP/IP uses physical, logical, port, and specific addresses for different purposes.